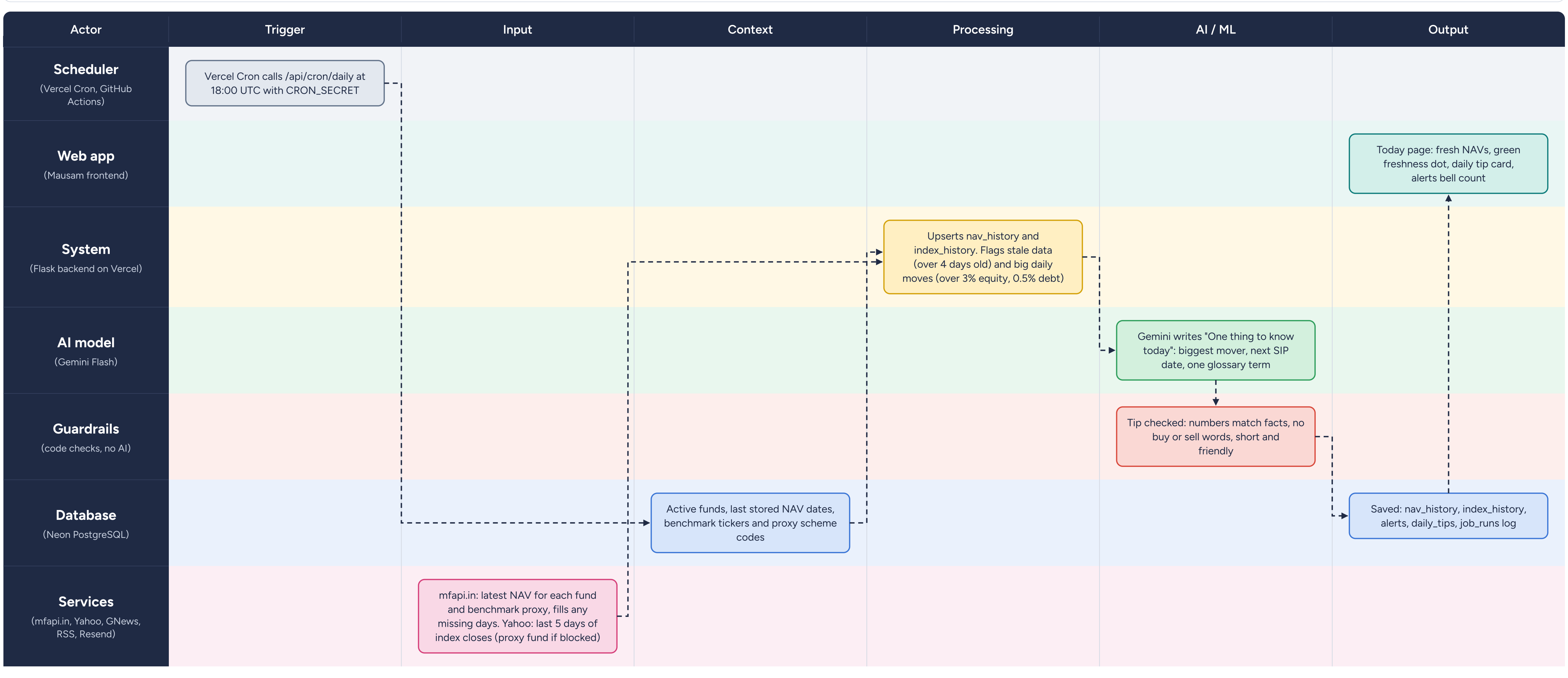
Mausam · Mutual Fund Weather Tracker — Components Map

Meera's five SIPs vs benchmarks and category averages, explained every Saturday by Gemini Flash, with a hybrid ARIMA + Ridge forecast range

Flow 1: Daily NAV job

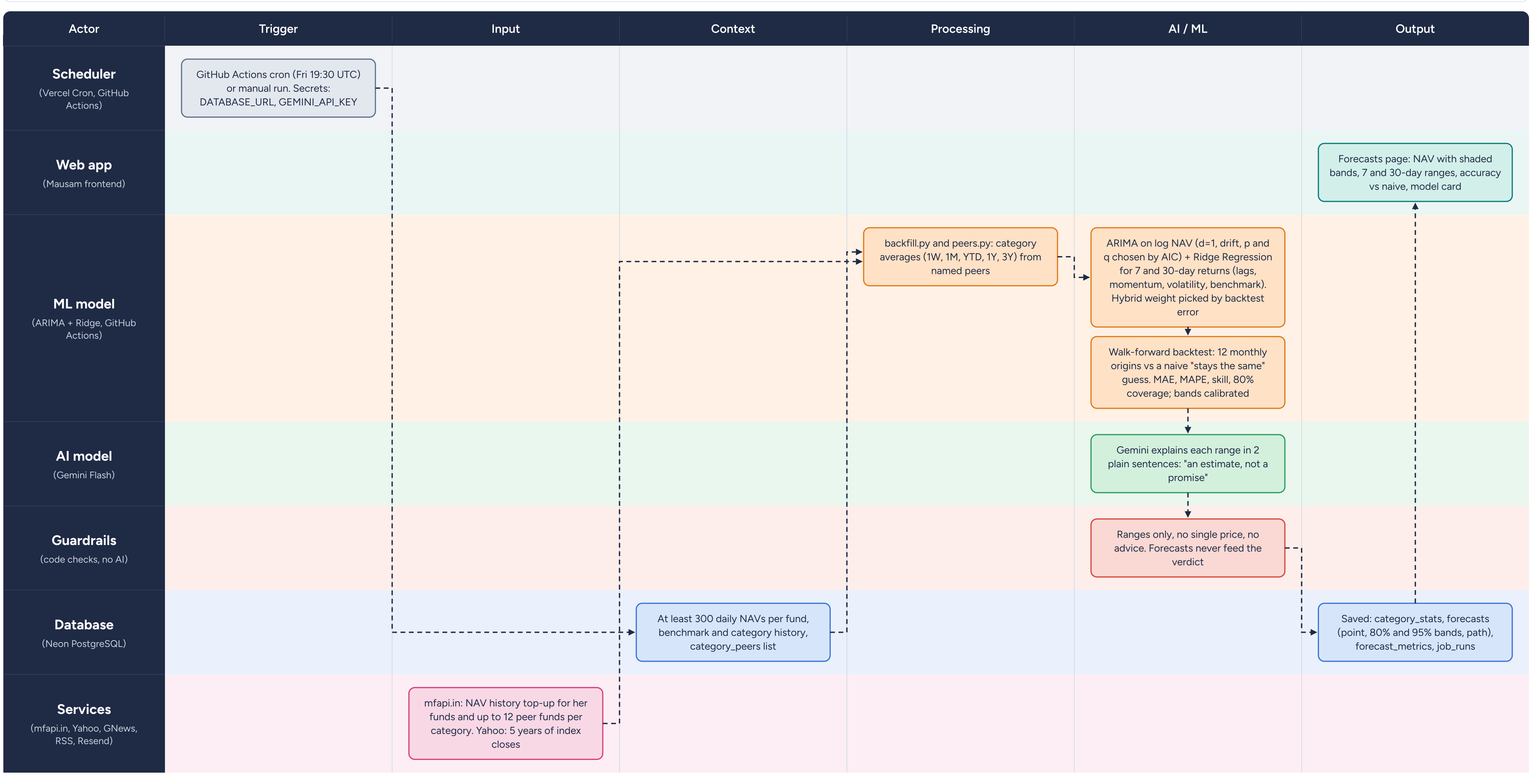
Every evening, about 23:30 IST



Actor	Trigger	Input	Context	Processing	AI / ML	Output
Scheduler (Vercel Cron, GitHub Actions)	Vercel Cron calls /api/cron/daily at 18:00 UTC with CRON_SECRET					
Web app (Mausam frontend)						Today page: fresh NAVs, green freshness dot, daily tip card, alerts bell count
System (Flask backend on Vercel)				Upserts nav_history and index_history. Flags stale data (over 4 days old) and big daily moves (over 3% equity, 0.5% debt)		
AI model (Gemini Flash)					Gemini writes "One thing to know today": biggest mover, next SIP date, one glossary term	
Guardrails (code checks, no AI)					Tip checked: numbers match facts, no buy or sell words, short and friendly	
Database (Neon PostgreSQL)			Active funds, last stored NAV dates, benchmark tickers and proxy scheme codes			Saved: nav_history, index_history, alerts, daily_tips, job_runs log
Services (mfapi.in, Yahoo, GNews, RSS, Resend)		mfapi.in: latest NAV for each fund and benchmark proxy, fills any missing days. Yahoo: last 5 days of index closes (proxy fund if blocked)				

Flow 2: Friday night forecasts and category averages

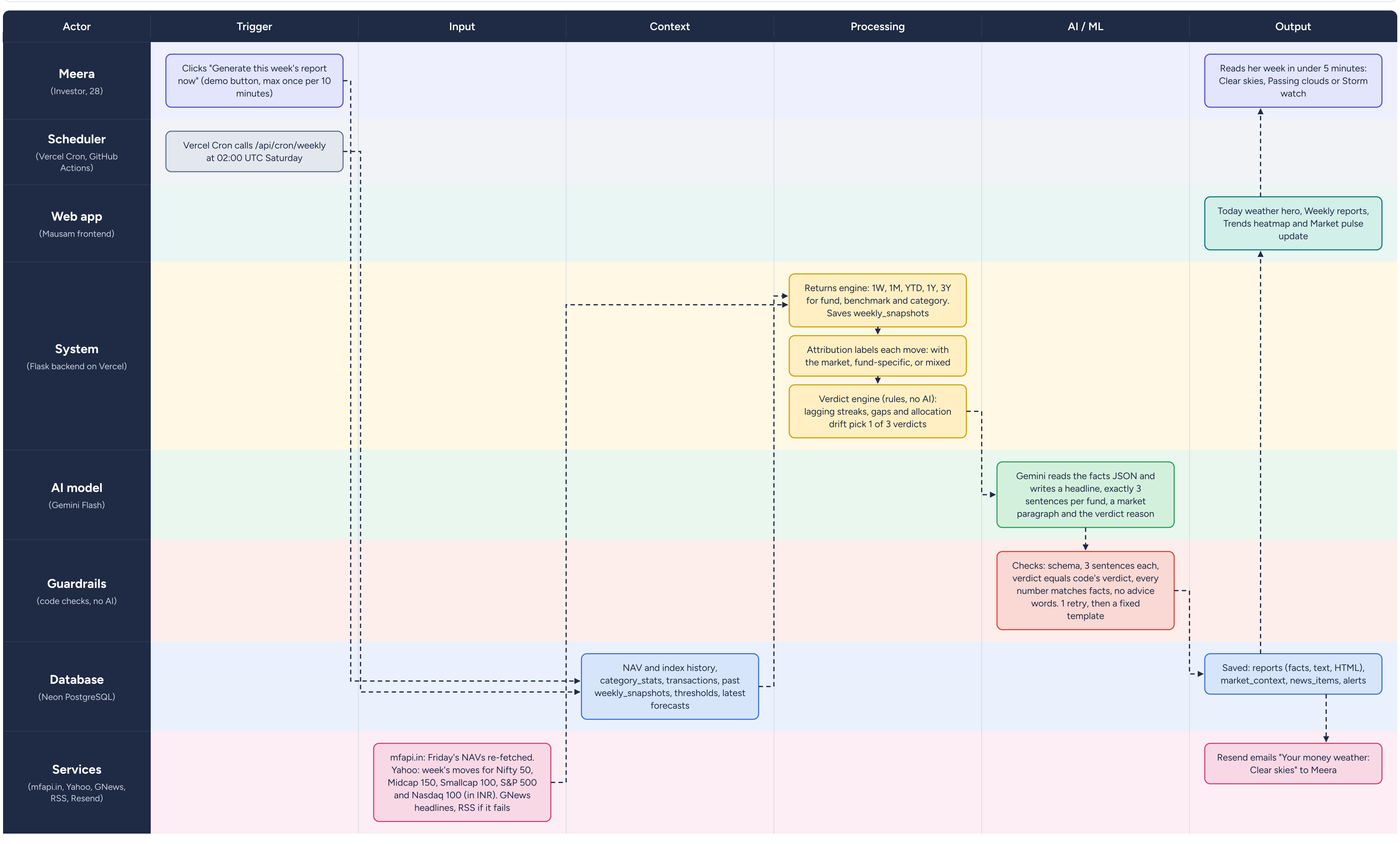
GitHub Actions, about 01:00 IST Saturday, or run manually



Actor	Trigger	Input	Context	Processing	AI / ML	Output
Scheduler (Vercel Cron, GitHub Actions)	GitHub Actions cron (Fri 19:30 UTC) or manual run. Secrets: DATABASE_URL, GEMINI_API_KEY					
Web app (Mausam frontend)						Forecasts page: NAV with shaded bands, 7 and 30-day ranges, accuracy vs naive, model card
ML model (ARIMA + Ridge, GitHub Actions)				backfill.py and peers.py: category averages (1W, 1M, YTD, 1Y, 3Y) from named peers	ARIMA on log NAV (d=1, drift, p and q chosen by AIC) + Ridge Regression for 7 and 30-day returns (lags, momentum, volatility, benchmark). Hybrid weight picked by backtest error Walk-forward backtest: 12 monthly origins vs a naive "stays the same" guess. MAE, MAPE, skill, 80% coverage; bands calibrated	
AI model (Gemini Flash)					Gemini explains each range in 2 plain sentences: "an estimate, not a promise"	
Guardrails (code checks, no AI)					Ranges only, no single price, no advice. Forecasts never feed the verdict	
Database (Neon PostgreSQL)			At least 300 daily NAVs per fund, benchmark and category history, category_peers list			Saved: category_stats, forecasts (point, 80% and 95% bands, path), forecast_metrics, job_runs
Services (mfapi.in, Yahoo, GNews, RSS, Resend)		mfapi.in: NAV history top-up for her funds and up to 12 peer funds per category. Yahoo: 5 years of index closes				

Flow 3: Saturday weather report

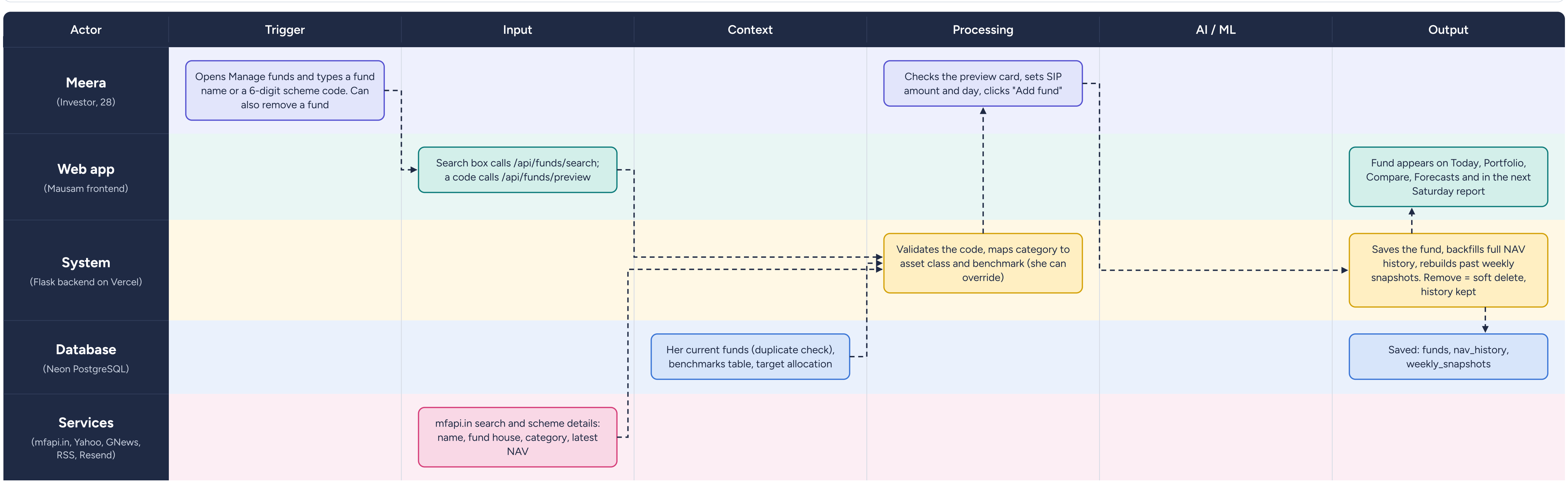
Vercel Cron about 07:30 IST Saturday, or the "Generate now" button



Actor	Trigger	Input	Context	Processing	AI / ML	Output
Meera (Investor, 28)	Clicks "Generate this week's report now" (demo button, max once per 10 minutes)					Reads her week in under 5 minutes: Clear skies, Passing clouds or Storm watch
Scheduler (Vercel Cron, GitHub Actions)	Vercel Cron calls /api/cron/weekly at 02:00 UTC Saturday					
Web app (Mausam frontend)						Today weather hero, Weekly reports, Trends heatmap and Market pulse update
System (Flask backend on Vercel)				Returns engine: 1W, 1M, YTD, 1Y, 3Y for fund, benchmark and category. Saves weekly_snapshots Attribution labels each move: with the market, fund-specific, or mixed Verdict engine (rules, no AI): lagging streaks, gaps and allocation drift pick 1 of 3 verdicts		
AI model (Gemini Flash)					Gemini reads the facts JSON and writes a headline, exactly 3 sentences per fund, a market paragraph and the verdict reason	
Guardrails (code checks, no AI)					Checks: schema, 3 sentences each, verdict equals code's verdict, every number matches facts, no advice words. 1 retry, then a fixed template	
Database (Neon PostgreSQL)			NAV and index history, category_stats, transactions, past weekly_snapshots, thresholds, latest forecasts			Saved: reports (facts, text, HTML), market_context, news_items, alerts
Services (mfapi.in, Yahoo, GNews, RSS, Resend)		mfapi.in: Friday's NAVs re-fetched. Yahoo: week's moves for Nifty 50, Midcap 150, Smallcap 100, S&P 500 and Nasdaq 100 (in INR). GNews headlines, RSS if it fails				Resend emails "Your money weather: Clear skies" to Meera

Flow 4: Meera adds or removes a fund

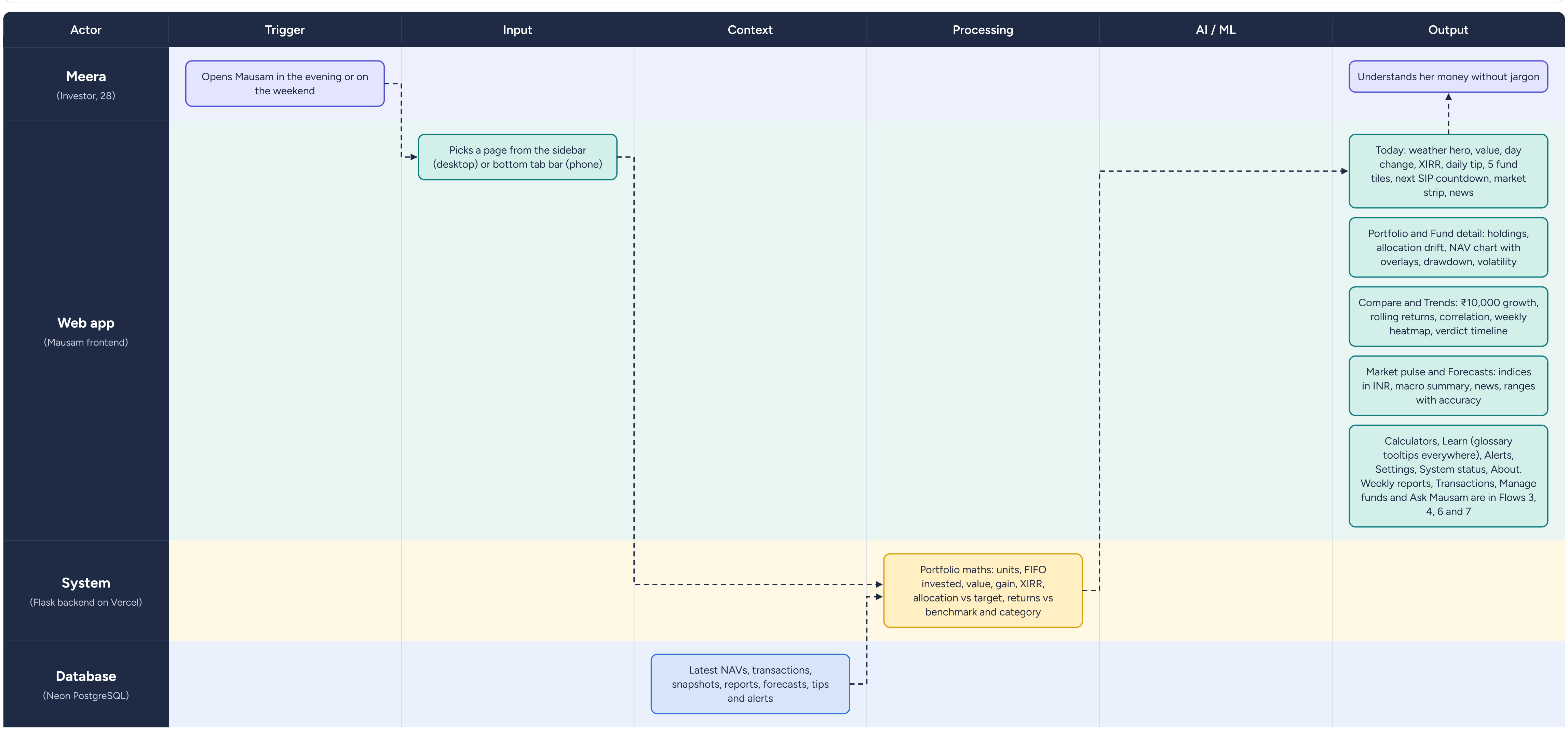
Manage funds page, no code changes



Actor	Trigger	Input	Context	Processing	AI / ML	Output
Meera (Investor, 28)	Opens Manage funds and types a fund name or a 6-digit scheme code. Can also remove a fund			Checks the preview card, sets SIP amount and day, clicks "Add fund"		
Web app (Mausam frontend)		Search box calls /api/funds/search; a code calls /api/funds/preview				Fund appears on Today, Portfolio, Compare, Forecasts and in the next Saturday report
System (Flask backend on Vercel)				Validates the code, maps category to asset class and benchmark (she can override)		Saves the fund, backfills full NAV history, rebuilds past weekly snapshots. Remove = soft delete, history kept
Database (Neon PostgreSQL)			Her current funds (duplicate check), benchmarks table, target allocation			Saved: funds, nav_history, weekly_snapshots
Services (mfapi.in, Yahoo, GNews, RSS, Resend)		mfapi.in search and scheme details: name, fund house, category, latest NAV				

Flow 5: Meera's daily visit

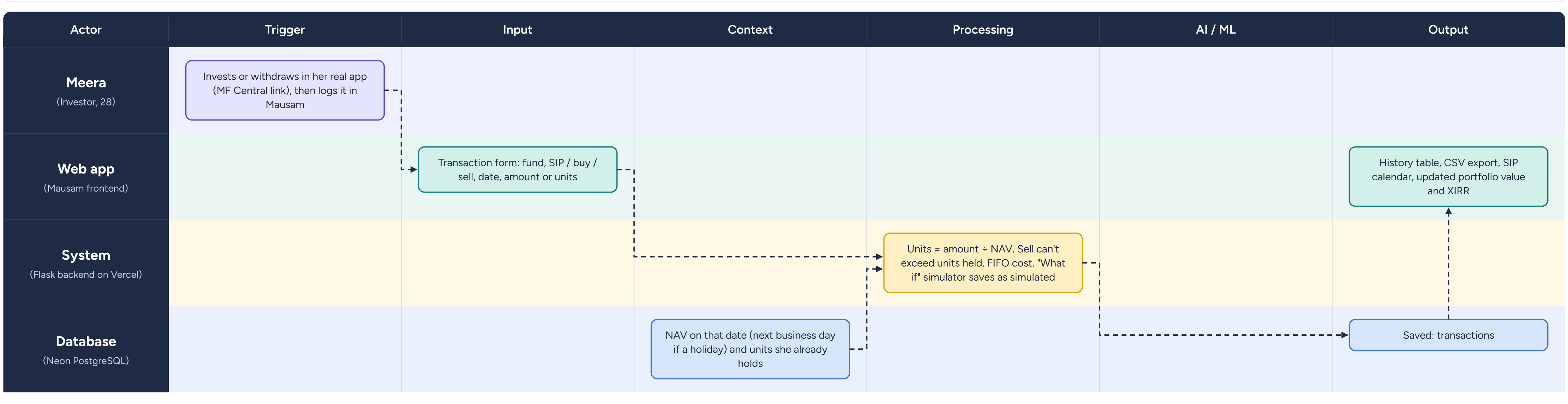
Any time, phone or laptop



Actor	Trigger	Input	Context	Processing	AI / ML	Output
Meera (Investor, 28)	Opens Mausam in the evening or on the weekend					Understands her money without jargon
Web app (Mausam frontend)		Picks a page from the sidebar (desktop) or bottom tab bar (phone)				Today: weather hero, value, day change, XIRR, daily tip, 5 fund tiles, next SIP countdown, market strip, news Portfolio and Fund detail: holdings, allocation drift, NAV chart with overlays, drawdown, volatility Compare and Trends: ₹10,000 growth, rolling returns, correlation, weekly heatmap, verdict timeline Market pulse and Forecasts: indices in INR, macro summary, news, ranges with accuracy Calculators, Learn (glossary tooltips everywhere), Alerts, Settings, System status, About. Weekly reports, Transactions, Manage funds and Ask Mausam are in Flows 3, 4, 6 and 7
System (Flask backend on Vercel)				Portfolio maths: units, FIFO invested, value, gain, XIRR, allocation vs target, returns vs benchmark and category		
Database (Neon PostgreSQL)			Latest NAVs, transactions, snapshots, reports, forecasts, tips and alerts			

Flow 6: Logging a buy, sell or SIP

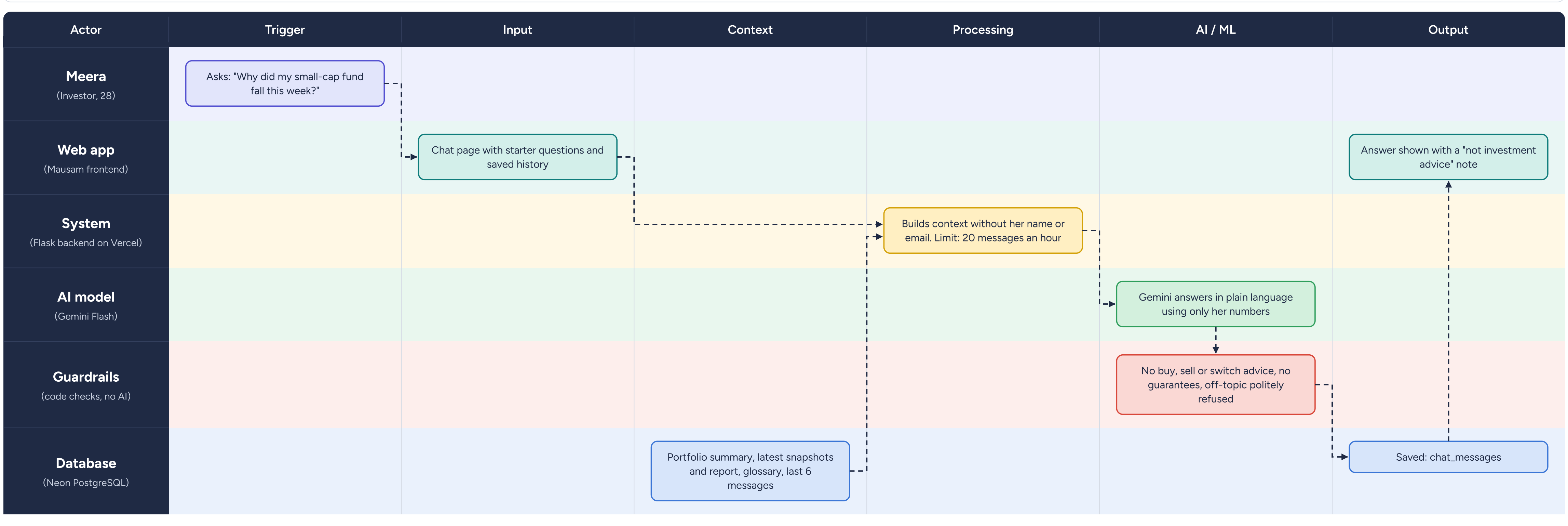
Transactions page (records only, never moves money)



Actor	Trigger	Input	Context	Processing	AI / ML	Output
Meera (Investor, 28)	Invests or withdraws in her real app (MF Central link), then logs it in Mausam					
Web app (Mausam frontend)		Transaction form: fund, SIP / buy / sell, date, amount or units				History table, CSV export, SIP calendar, updated portfolio value and XIRR
System (Flask backend on Vercel)				Units = amount ÷ NAV. Sell can't exceed units held. FIFO cost. "What if" simulator saves as simulated		
Database (Neon PostgreSQL)			NAV on that date (next business day if a holiday) and units she already holds			Saved: transactions

Flow 7: Ask Mausam (chat)

Chat page



Actor	Trigger	Input	Context	Processing	AI / ML	Output
Meera (Investor, 28)	Asks: "Why did my small-cap fund fall this week?"					
Web app (Mausam frontend)		Chat page with starter questions and saved history				Answer shown with a "not investment advice" note
System (Flask backend on Vercel)				Builds context without her name or email. Limit: 20 messages an hour		
AI model (Gemini Flash)					Gemini answers in plain language using only her numbers	
Guardrails (code checks, no AI)					No buy, sell or switch advice, no guarantees, off-topic politely refused	
Database (Neon PostgreSQL)			Portfolio summary, latest snapshots and report, glossary, last 6 messages			Saved: chat_messages

Reference

Database tables (Neon)

- users, settings: email, risk profile, target allocation, thresholds
- funds: scheme code, category, asset class, benchmark, SIP, active
- nav_history: every daily NAV per scheme
- benchmarks, index_history: tickers, proxies, daily closes
- category_peers, category_stats: peers and weekly averages
- transactions: SIP, buy, sell, NAV used, units, simulated flag
- weekly_snapshots: every fund every Friday
- reports: verdict, facts, Gemini text, HTML, emailed
- market_context, news_items: index moves, headlines
- forecasts, forecast_metrics: ranges, skill, coverage
- alerts, daily_tips, chat_messages
- job_runs: status, timing, fallbacks for every job

Guardrails and risks

- AI invents numbers: maths in code, number check on every output
- AI invents news: news cited by ID only
- AI gives advice: banned phrases, fixed 3 verdicts, disclaimers
- Forecast taken as a promise: ranges and accuracy vs naive
- Data late: retries, stale badges, Saturday re-fetch
- Yahoo blocked: index-fund proxy, labelled in the app
- News API fails: RSS fallback, "no major news" wording
- Wrong benchmark: category mapping + manual override
- Category average unofficial: shows peer count
- Missed job: job_runs log, status page, "Generate now"
- Secrets: env vars only, CRON_SECRET on job routes
- Privacy: no name or email sent to Gemini

Services, keys and accounts

- mfapi.in: NAVs and scheme details, no key
- Yahoo Finance chart: index closes, no key (unofficial)
- GNews: news, GNEWS_API_KEY (RSS as backup)
- Google AI Studio: GEMINI_API_KEY, gemini-2.5-flash
- Neon: DATABASE_URL (pooled)
- Resend: RESEND_API_KEY, REPORT_EMAIL_TO
- Vercel: hosting, Flask function, 2 cron jobs
- GitHub: code, Actions for ML and CI, secrets
- CRON_SECRET: protects the job routes
- Not used: NewsAPI.org (localhost only), Claude API
- Not possible: real buy or sell orders
- All free tiers: ₹0 a month

Nine checks

- 01 Problem real: yes, monthly screenshots to a CA
- 02 Repeated: yes, every weekend
- 03 Input: partial, no free benchmark or trading API
- 04 Output: yes, plain report + one verdict
- 05 Measurable: yes, on-time reports, forecast skill
- 06 Failure risk: yes, with guardrails
- 07 Judgment: yes, explains and never transacts
- 08 ROI: yes, ₹0 a month
- 09 Owner: yes, Meera owns funds and decisions
- The cut: real buy/sell orders
- The cut: single-number price predictions
- Verdicts: Clear skies, Passing clouds, Storm watch

MESA Cohort C4, AI and its Applications, Section 1 final. Educational tool, not investment advice. Mutual fund investments are subject to market risks.